

# Brendon Franz Matusch

## Contact Information

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## Overview

I am a Master's student in Mechanical Engineering at the National University of Singapore, and graduated in Computer Science at Stanford University. I have a wide range of experience in AI, control techniques, and embedded robot programming. I enjoy working on all aspects of robotics and automation: hardware, software, and the interface between them.

I have worked on natural language processing research at Midjourney, on physically realistic robot simulations at NVIDIA, on autonomous food delivery robots at HYPR, and on home backup battery systems at Base Power Company.

## Education

### ***National University of Singapore, Master's (Aug 2025 to May 2026)***

- Working towards Master of Science in Mechanical Engineering.

### ***Stanford University, Undergraduate (Sep 2020 to Dec 2024)***

- Working towards Bachelor of Science in Computer Science (under the School of Engineering) with a specialization in Artificial Intelligence.
- Cumulative GPA of 3.895.
- Graduated in 10 academic quarters (3 ½ academic years); took one quarter of leave to work at NVIDIA and two quarters of leave to work at HYPR.

## Employment and Research History

### ***Intern at Base Power Company (Austin, TX, Jun to Aug 2024)***

- Base Power Company manufactures home backup battery systems, and controls the fleet of batteries to reduce energy costs for the customer and stabilize the electrical grid. I worked on software to track the performance of the fleet and determine sources of energy losses in hardware.
- I also worked on the embedded and cloud software used to reliably remotely control the fleet.

### ***Senior Project at Stanford University with Mercedes-Benz (Jan to Jun 2024)***

- Worked with four other Stanford students to design and build a research application, reporting to engineers at Mercedes-Benz Research and Development North America.
- Developed a system for automatic road sign language translation and parking sign interpretation.
- Presented work at Mercedes R&D headquarters in Stuttgart, Germany.

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***Intern at HYPRLABS (HYPR), working on autonomous food delivery robots (Paris, France, part-time Sep to Dec 2022 and full-time Jan to Jun 2023)***

- HYPR develops delivery robots that can navigate crowded urban areas. I led the effort to design and build a subsystem to identify possible crossing points at intersections and determine when and where to safely cross the road.
- I also worked on human-robot interactions, neural network explainability, optimizing compute efficiency for onboard hardware, mapping interfaces for navigation, and improving autonomous driving functionality.

***Intern at NVIDIA, working on Isaac Gym simulation technology and reinforcement learning (Mar to Sep 2021)***

- I worked on developing physically realistic, parallelized, and GPU-accelerated 3D robotics simulations.
- The software is used for reinforcement learning and simulation-to-real-world transfer applications.

***Founding AI researcher at Midjourney, working on natural language experiments (Jan to Aug 2020)***

- Developed an AI application that provides the ability to explore a text repository in the style of a mapping application. The user can zoom in and out in abstract information space, where nearby articles or documents are neighboring, and topic areas are labeled as regions on the map.
- Worked on early experimental diffusion for image generation, and several other experiments.

***Lead researcher on unsupervised reinforcement learning project at Vector Institute (May 2019 to Jan 2020)***

- Studied and implemented potential exploration objectives for unsupervised reinforcement learning, and statistically evaluated their empirical effectiveness on agent behavior.

***Independent student researcher on machine learning techniques to accelerate dark matter research at SNOLAB (Sudbury, ON, May 2018 to May 2019)***

- Independently developed models to distinguish potential dark matter candidates, specifically weakly-interacting massive particles, from background alpha radiation of radioactive-decay origin.
- Developed approaches specific to two dark matter detection experiments: PICO-60 and DEAP-3600.

***Converted an electric go-kart to a self-driving vehicle (May 2017 to May 2018)***

- Mechanically/electrically modified a go-kart for autonomous steering and cruise control.
- Developed and compared multiple machine learning models for road identification, and control algorithms to steer the vehicle. Software was built from scratch and trained using data that I collected and labeled.

## ***Awards and Achievements***

### ***Neo Scholar, 2023 to 2024***

- Selected as one of 32 scholars among 1000 applicants by Neo, an established venture capital firm, based on entrepreneurial potential and technical excellence.

### ***Masason Foundation Scholar, 2021 to 2026***

- The Masason Foundation was founded by Masayoshi Son, CEO of SoftBank Group, to support students with exceptional talent in engineering and the sciences.
- Selected as one of 31 scholars in 2021 for a full-tuition university scholarship, and financial support for research and business endeavors.

### ***Associate Diploma (ARCT) in Piano Performance from the Royal Conservatory of Music, 2021***

- The ARCT is the second-highest academic standing awarded by the Royal Conservatory, awarded to 375 students in North America in 2021, among approximately 500,000 students studying under the Royal Conservatory curriculum.
- Earned after 14 years of practice in piano performance, and study in music theory and history.

### ***Best Project at Canada-Wide Science Fair (CWSF), and First Prize at the European Union Contest for Young Scientists, 2018***

- Autonomous driving project was selected as the top science/engineering project in Canada, and one of the top three projects at EUCYS, hosted by the EU and including competitors from around the globe.

### ***Conference Presentations and Other Awards***

- Presented senior project at Stanford University at the Software Fair for 75 student projects, and was selected for one of three awards sponsored by Google and one of three sponsored by Meta.
- Reinforcement learning research was selected for an oral presentation at Biological and Artificial RL Workshop, and selected for the Deep RL Workshop, both at NeurIPS 2020.
- Research on dark matter detection was selected for poster presentation at CERN's Advanced Computing and Analysis Techniques conference in Saas Fee, Switzerland, and at NVIDIA's GPU Technology Conference in San Jose, California, in 2019.
- Second Award at the Intel International Science and Engineering Fair (ISEF) in Phoenix, Arizona, 2019.
- Top 100 projects globally at the Google Science Fair in 2019.